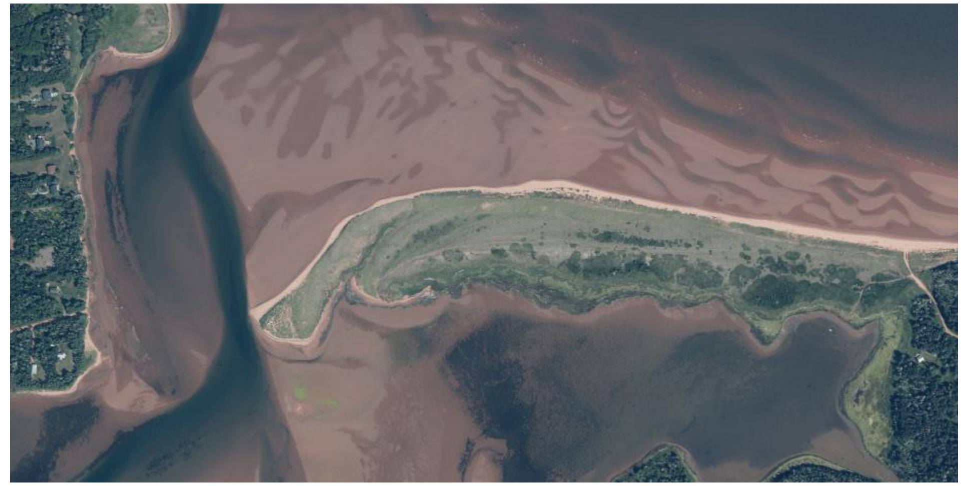
Howe Bay

Shoreline length: 23 km



Barrier sandspit at Howe Bay entrance

Shore Types

Howe Bay SMU includes an open coastline facing East and an estuary protected by a barrier sandspit, with a majority of low plains, cliffs, and wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
40	5	0	29	21	0	0	5

Head of Howe Bay estuary

Governance, Infrastructure, and Economy

Approximately 43% of this SMU is unincorporated; the remaining 57% lies within the Rural Municipality of Annandale-Little Pond-Howe Bay.

Coastal infrastructure includes Grams Beach Road towards the sandspit, Chads Road at the beach end on the north side, and Route 310. There are no significant coastal economic drivers identified. Local and legacy land use constraints include few small lots on private roads along the coast.

Natural Assets and Heritage

There are no Provincial or Federal Parks, nor Wildlife Management Areas, in this SMU. The shoreline includes critical habitat for bird species including the endangered Piping Plover and threatened Bank Swallow, notably along the outer cliffs and the sand spit.

Planning and Policy Framework

Both the municipal and unincorporated areas in this SMU fall under provincial planning authority. Provincial rules apply throughout this SMU. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions.

There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Coastal Flood and Erosion Risk

While low plains and wetland shores are vulnerable to increasing flood hazards, infrastructure risk is low. The erosion hazard is highest along the exposed east shoreline. Erosion hazard inside the estuary is limited by the sheltered exposure and barrier sandspit.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline]

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
SMU Howe Bay				
Flood risk	0	1 [<1/km]	2 [<1/km]	3 [<1/km]
Erosion risk	N/A	2 [<1/km]	4 [<1/km]	13 [1/km]
PU Head of Estuary				
Flood risk	0	0	0	1 [<1/km]
Erosion risk	N/A	1 [2/km]	1 [2/km]	2 [3/km]

Sediment transport - Sediment transport potential is highest along the exposed cliffs/ bluffs, which feed the barrier spit.

Proposed SMP Strategies

It is critical to maintain sediment transport towards the sandspit protecting the bay. Given the low risk levels the proposed strategy is NAI across the three epochs, except for the head of the estuary where MR would be warranted notably for the road due to flood risk.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	**NAI**	**NAI**	**NAI**
Rationale	The local shoreline and hazard zone should remain natural, undeveloped.		
Proposed PU strategy **Head of Estuary**	**MR**	**MR**	**NAI**
Rationale	MR will reduce risk to the limited number of vulnerable assets and the need for future intervention.		

LEGEND

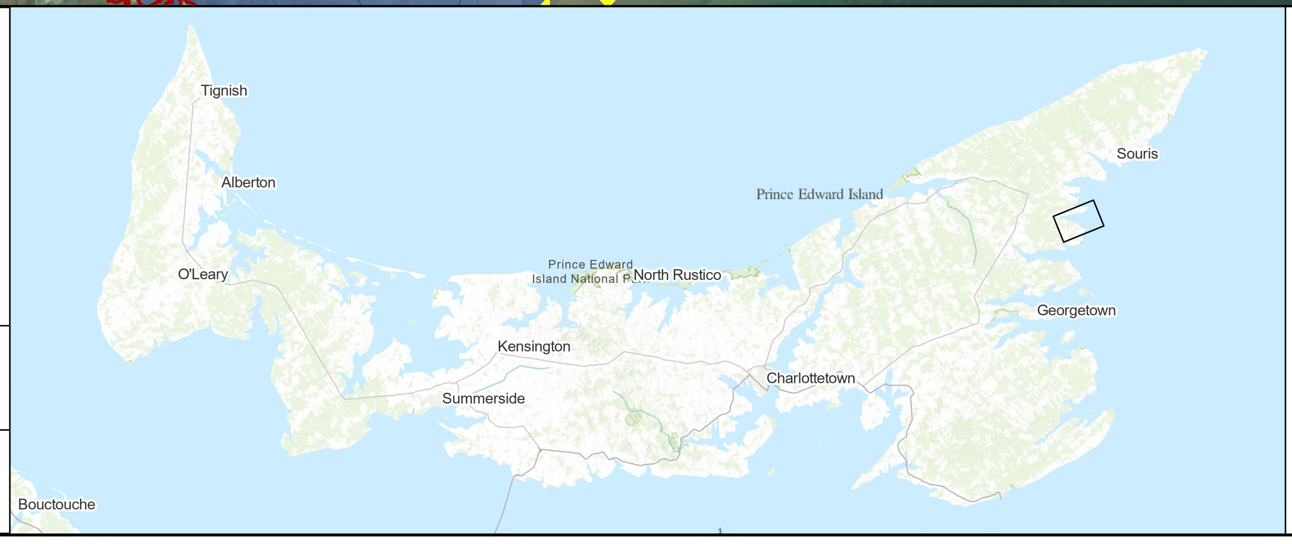
- Policy Unit
- Shoreline Management Unit
- Net Longshore Sediment Transport
- High Hazard Area
- Flood Extent - Present Day (100-year Return)
- Flood Extent - 1.2m SLR (100-year Return)
- Property Boundary
- Building > 90ft²

N W E S

0 115 230 460
Metres

Coordinate System:
NAD 83 CSRS PEI Double Stereographic

NOTES:



COASTAL EROSION AND FLOOD HAZARD

Shoreline Management Unit: Howe Bay

Prince Edward Island
Île-du-Prince-Édouard

CBCL